

Amira Mouamine

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EDUCATION

Université de Montréal

M.Sc. in Computer Science, Specialization in Artificial Intelligence

Montreal, QC

Expected 05/28

Université Laval

B.Eng. in Computer Engineering

Quebec, QC

09/22 – 05/26

- **Relevant Coursework:** Game Development Programming, Computer Graphics, Operating Systems, Real-Time Embedded Systems, Machine Learning, Database Models and Languages

TECHNICAL SKILLS

Programming Languages: C++, C#, C, Python, JavaScript/TypeScript, Java, Rust, GLSL, SQL, VHDL, MATLAB

Frameworks & Libraries: OpenGL, openFrameworks, OpenCV, React, Node.js, scikit-learn, PyTorch, TensorFlow

Tools: Git, Azure DevOps, Docker, Azure Blob Storage, MySQL, Unity, Unreal Engine, Blender, Xilinx Vivado

EXPERIENCE

Bentley Systems

05/25 – 08/25

Full Stack Developer Intern

- Unblocked and shipped a **long-requested auto-save measurement feature** across Bentley civil design products, coordinating refactors across repositories and services for **cross-product consistency**.
- Built with **TypeScript, React, and iTwin.js**, the feature persisted manual measurements across sessions through create, rename, duplicate, delete, and recover workflows, **reducing data loss and rework**.
- Integrated secure cloud storage with **Azure Blob Storage and SAS tokens**, handling partial saves and failure cases through targeted testing and debugging.

PROJECTS

Graphics Editor | C++, GLSL, OpenGL, openFrameworks

- Built a **real-time 2D/3D graphics editor** in C++/OpenGL with **6 GLSL shading models**, PBR materials, normal mapping, multi-light rendering, HDR skybox and FBO tone mapping.
- Implemented interactive scene authoring with runtime shader switching, editable **16-point bicubic Bézier surfaces** with GPU tessellation, camera controls and image/3D model import/export.

EMG Gesture Recognition for F1TENTH | Python, scikit-learn

- Designed a **signal-processing and evaluation pipeline** for subject-specific EMG classification in an **applied research project**, after finding that pooled participant data masked meaningful individual differences.
- Achieved **89.6% accuracy** on held-out recordings and deployed the classifier for real-time control of an **F1TENTH vehicle**, earning the **Coup de Cœur Award** and the **Jonathan Gaudreault Scholarship**.

Operations Management Platform for Les Narrateurs | Python, Flask, MySQL, Vue.js

- Consolidated years of fragmented Excel records into a normalized **MySQL relational database**, identifying inconsistencies and duplicates to establish reliable operational data for a **Québec nonprofit**.
- Built **Flask REST APIs** with parameterized SQL and role-based access for contracts, clients, schedules, activities and reports, enabling staff to **generate consistent reports without manually reconciling spreadsheets**.

Cat & Rats Game | C#, Unity, Aseprite

- Built a **two-player co-op game** with remappable controls, procedural levels and multi-scene flow.
- Implemented an **A* path-finding cat enemy** with state-based behavior reacting dynamically to both players.
- Added **avatar customization, minimap, event-driven animations, VFX, SFX** and dynamic music.

ACTIVITIES & LEADERSHIP

Game Development Club of Université Laval

09/24 – 05/26

Co-Leader

- Organized game jams, industry talks and technical workshops for a community of approximately **30 members**.

Artificial Intelligence Club of Université Laval

09/25 – 05/26

Member

- Competed in the **JFFI hackathon**, solving technical challenges collaboratively under time constraints.